StaticKD: Distilling a Multilingual IPTC News Topic Classifier into a Static Token-Logit Table for Lightweight CPU Inference

Obi Kastanya and Ananta Dwi Prayoga Alwy

***Abstract*—Filtering news by the International Press Telecommunications Council (IPTC) Media Topic taxonomy needs a classifier cheap enough for a multilingual stream of articles. The open multilingual IPTC classifier, based on XLM-RoBERTa-large, takes 1,239 to 2,971 milliseconds per article on one processor core. StaticKD uses this classifier as a teacher and stores 17 scores for each vocabulary token, one per top-level IPTC topic. Classifying an article then only requires tokenizing it and averaging its token scores. The student starts from multilingual static embeddings (one fixed vector per token) and learns from teacher outputs on 285,999 articles in 69 languages. A new test set of 12,744 articles in 47 languages uses topics assigned by human annotators or newsroom editors and mapped onto IPTC, so no test label comes from a model that reads the article. On this test set StaticKD reaches a macro-F1 of 0.545, or 91.1% of the teacher score of 0.598. It classifies 473 articles per second per core, about 616 times faster than the 8-bit quantized teacher, with a 34-megabyte model, and is more accurate than fastText and a linear support vector machine trained on the same teacher labels. Ablations with five seeds show that accuracy comes mainly from multilingual training data and from pre-trained cross-lingual embeddings, and that the top teacher label works as well as the full distribution. Fine-tuning the embeddings helps the training languages but hurts unseen languages. StaticKD is weakest on languages absent from both its training data and the teacher's pre-training data.**

***Index Terms*—Efficient inference, knowledge distillation, multilingual text classification, news topic classification, static embeddings.**

I. Introduction

NEWS analytics pipelines process a continuous stream of articles from many sources and languages. Many of their later stages, such as entity extraction, sentiment scoring, or event detection, are needed only for some topics. Topic classification therefore comes first and decides which articles each later stage receives, which makes it the only stage that reads every incoming article. Its cost grows with the whole stream, whereas the cost of the later stages grows only with the articles that pass the filter. A topic classifier at the head of a pipeline thus has to be cheap per article and has to work in every language of the stream. Routing by topic also needs the same topics for all sources, while publishers name their sections in different ways; the mapping table built in this study alone covers 814 section names (Section III-D). The taxonomy most used by news agencies is the Media Topic taxonomy of the International Press Telecommunications Council (IPTC). Its top level has 17 topics, such as *politics*, *economy, business and finance*, and *sport* [1], [2].

Kuzman and Ljubešić [2] released the first open multilingual IPTC classifier. Articles in four languages were labeled with GPT-4o, and these labels were used to fine-tune XLM-RoBERTa-large [3]. The resulting model matches GPT-4o on a human-annotated test set. A follow-up study by the same group concluded that fine-tuned encoders are more practical than large language models (LLMs) for large-scale annotation [4]. An encoder with 559.9 million parameters is still expensive without a graphics processing unit (GPU), and this study targets servers without one, where the classifier shares central processing unit (CPU) cores with the rest of the pipeline. On one CPU core the model needs 2,971 ms per article, and its fastest CPU configuration, an Open Neural Network Exchange (ONNX) export quantized to 8-bit integers (int8), still needs 1,239 ms. A stream of one million articles per day would then occupy 15.1 (ONNX int8) to 34.8 (PyTorch) CPU cores for topic classification alone (Section IV-D). Neither study reports this inference cost, although inference accounts for a large share of the computing demand and energy use of machine learning in production, for example 60% of the machine learning energy use reported by Google [5].

Such a pipeline needs a topic classifier that covers many languages, runs in milliseconds per article on an ordinary CPU, and has a small footprint and few dependencies. Small distilled transformers [6], [7] gain about one order of magnitude. In the CPU benchmark of this study, MiniLM-L6 is 21 times faster than the ONNX int8 teacher. Int8 quantization alone gives a factor of 2.3 over PyTorch fp32. fastText [8] is much faster, but it learns its word vectors from scratch and therefore does not transfer across languages. This is a serious restriction here. The IPTC training labels released with the open classifier cover only four languages [9], and even a corpus labeled automatically by a multilingual teacher cannot contain every language of a news stream. No available IPTC classifier is both multilingual and this cheap.

This paper proposes StaticKD, which distills [10] the XLM-R-large IPTC classifier into a far simpler student. Three properties of the problem make this possible. A teacher that works across languages can label unlabeled news in any language, so training data for the student cost no annotation. Multilingual static embeddings, distilled from sentence encoders [11], [12], [13], give each token one fixed vector and place translation equivalents close to each other, and looking up such a vector

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costs almost nothing. News topics are also strongly tied to vocabulary, which is why bag-of-words representations remain competitive for topic classification [14]. The student therefore averages the embeddings of the tokens in an article and applies one linear layer. Since both steps are linear, the linear layer can be applied to every embedding once, before deployment. This paper calls this step folding. It turns the model into a table that stores 17 topic scores (logits) per token. Classifying an article then reduces to tokenizing it and averaging the table rows of its tokens, which NumPy handles without a deep learning library. Fig. 6 in Section IV-D shows where the resulting model lies in accuracy and speed relative to the other models.

A multilingual claim also requires a multilingual evaluation with labels that do not come from the teacher. The human test set of Kuzman and Ljubešić is not public and covers four languages. SIB-200 [15] covers 205 languages, but its items are short sentences with a non-IPTC taxonomy. To fill this gap, a unified test set of 12,744 articles in 47 languages is built. It combines news pages from 2024 to 2026 whose publishers record the newsroom section of the article in the page metadata (schema.org [16]) with public datasets labeled by human annotators or with publisher categories [17], [18], [19], [20]. These labels are editorial decisions of publishers or annotators, translated onto the official IPTC taxonomy. No model that reads the article text is involved.

The study answers four research questions (RQs). RQ1 asks how much of the teacher accuracy StaticKD retains on labels that do not come from the teacher. The comparison includes two baselines of similar cost, fastText and a support vector machine (SVM) on term frequency–inverse document frequency (TF-IDF) features, and two more expensive small transformers, MiniLM-L6 and DistilmBERT. RQ2 examines which parts of the training recipe matter for this very small student. The parts tested are soft labels, multilingual data, embedding initialization, embedding fine-tuning, token dropout, temperature, and ensembling. RQ3 measures the CPU inference cost of the teacher and how far StaticKD reduces it. RQ4 looks for the languages, labels, and label sources on which StaticKD is weak, and for the factors behind these weaknesses.

The answers also show where the accuracy of such a small student comes from. What matters is how many languages and sites the teacher-labeled data cover, and whether the embeddings start from a space shared across languages. The exact probabilities that the teacher assigns to the other topics matter little. Training on the top teacher label alone works as well as training on the full teacher distribution, and a high distillation temperature hurts. Languages that never occur in the training data are classified through embeddings that training leaves largely unchanged, which works because these embeddings share one space with those of the trained languages. Fine-tuning the embeddings therefore helps the training languages but hurts the unseen ones, and a development set in the training languages cannot reveal this trade-off.

The contributions of this paper are as follows.

1) StaticKD, a recipe that distills a multilingual transformer teacher into a table of 17 logits per token, initialized from multilingual static embeddings. The final model takes 34 MB, needs only the `numpy` and `tokenizers` libraries, and processes 473 articles per second per core. An ensemble of several trained students folds into the same single table, so it adds no inference cost.
2) A multilingual IPTC test set of 12,744 articles in 47 languages whose editorial labels do not come from the teacher. It comes with a section-to-IPTC mapping table that is checked automatically and reviewed by the first author, deduplication across sources, a contamination check against all training data, and confidence intervals that account for articles from the same site being related. It can be reused to evaluate other IPTC classifiers.
3) The findings above on where the accuracy of the student comes from, supported by ablations with five seeds per variant and by analyses by language, label, label origin, and source dataset, including languages that are absent from the training data. The accuracy and cost of nine configurations from six model families, StaticKD included, are measured on CPU only, with each process pinned to the fast cores of a hybrid CPU.
4) A release of the code, a notebook that reproduces every number in this paper, the models, and the test-set manifest.

Section II reviews related work, Section III describes the data, test set, method, and evaluation protocol, Section IV presents the results, Section V discusses them, Section VI lists the limitations, and Section VII concludes.

II. Related Work

A. Multilingual News Topic Classification and the IPTC Scheme

Resources for news topic classification usually define their own taxonomy and cover a single language or region. MasakhaNEWS covers 16 languages widely spoken in Africa, English and French included, with seven human-annotated categories [17]. L3Cube-IndicNews uses publisher site categories for 10 Indian languages [19], and a Bangla news dataset also relies on publisher categories [20]. MN-DS is the only public dataset annotated directly with top-level IPTC topics, and it is English only [18]. SIB-200 spans 205 languages, but its items are Flores-200 sentences with seven non-IPTC topics [15]. Kuzman and Ljubešić [2] showed that GPT-4o labels can replace human annotation for training an IPTC classifier, and their follow-up work found fine-tuned encoders more practical than LLMs for large-scale annotation [4]. The only open multilingual IPTC classifier is an encoder with 559.9 million parameters, and no inexpensive IPTC classifier has been evaluated on real news in many languages.

B. Distilling Transformer Classifiers into Compact Students

Knowledge distillation (KD) trains a cheap student model to imitate the output distribution of an expensive teacher model [10], [21]. Both logit vectors are divided by a temperature τ before the softmax. A higher τ flattens the distribution, so the student also learns how probable the teacher finds the wrong classes. The full distribution is called a soft label, and the top class alone is called a hard label. In text classification the student is almost always a transformer,

for example DistilBERT [6] or MiniLMv2 [7], including cross-lingual distillation that moves teacher knowledge to languages with little data [22]. The closest prior work is Sparse Distillation [23], which distills RoBERTa-large into a deep averaging network [24] with a very large n-gram table and runs hundreds of times faster on English single-sentence classification tasks. StaticKD differs in three ways. Its table starts from multilingual static embeddings that are already aligned across languages instead of being trained from scratch. Its linear head is folded so that each token stores only 17 numbers. It targets news documents in 69 languages and is evaluated on languages that are missing from the distillation data.

C. Temperature and the Capacity Gap

The success of distillation depends on the capacity gap between teacher and student, which grows as the teacher becomes larger and more expressive than the student. For language models, the optimal teacher size even follows an empirical law with respect to the student size [25]. Temperature is treated as an important hyperparameter that can be learned [26]. The Kullback–Leibler (KL) divergence can be split into a target-class part and a non-target part (dark knowledge) [27], and recent analysis shows that KL under a capacity mismatch distorts the ranking of classes [28]. High agreement between student and teacher (fidelity) also does not always come with better generalization [29]. Nearly all of these findings come from deep students. How temperature and soft labels behave at an extreme gap, from 559.9 million parameters to one linear layer over averaged embeddings, has not been characterized.

D. Static Embeddings and Efficient CPU Inference

A static embedding gives each token one fixed vector, independent of the surrounding words, so a text vector is simply the mean of its token vectors. Model2Vec [11] and Wada et al. [12] extract such embeddings from sentence transformers, and their averages are competitive sentence embeddings. The potion-multilingual variant of Model2Vec is distilled from BGE-M3 [13] and covers 101 languages. The Model2Vec library also offers a classifier trainer, which adds a 512-unit hidden layer and learns from hard labels [11]. Static embeddings are usually evaluated as general-purpose embeddings, not as distillation students for a specific taxonomy. A survey of efficient natural language processing (NLP) methods recommends reporting inference cost next to accuracy [30], since inference makes up a large share of the energy use of deployed models [5]. Few studies, however, report CPU latency at low thread counts against a quantized teacher.

E. Test Sets with Editorial Labels, Contamination, and Sample Dependence

Publisher site categories have been used as topic labels before [19], usually for one publisher or one region. Many news sites also store the section of each article in machine-readable page metadata, the schema.org `articleSection` field [16]. This field has not yet been mapped to a standard taxonomy across dozens of languages. Leakage between training and test data also has to be controlled, and near-duplicate removal with MinHash [31] has become standard practice [32]. Significance tests in NLP such as the paired bootstrap [33] usually assume independent samples [34], while articles from the same site are dependent. The cluster bootstrap [35] handles this dependence by resampling clusters instead of documents. Table I summarizes how StaticKD relates to these approaches.

III. Methodology

A. Study Design

The study has three parts: building the StaticKD student by distillation, measuring its accuracy against labels that do not come from the teacher, and measuring its inference cost on CPU.

Fig. 1 shows the StaticKD pipeline built in the first part. The teacher reads 285,999 unlabeled news articles in 69 languages once, offline, and its 17 topic logits for each article are stored (Sections III-C and III-E). StaticKD is trained to reproduce the full teacher distribution over the 17 topics (Section III-F). The baselines learn from the same articles: the two small transformers from the full distribution, and fastText and TF-IDF + SVM from the top teacher label only. One further fastText model is trained on GPT-4o labels instead, as a reference without the teacher (Section III-G). After training, the linear head of StaticKD is folded into a table of 17 scores per token, and this table is the only part used at inference.

Scores against teacher labels show only how closely a model imitates the teacher, and every teacher error that a student copies would count as correct. The second part therefore uses a separate test set whose editorial labels come from newsroom sections, publisher categories, and human annotators (Section III-D). All accuracy comparisons, statistical tests, and subgroup analyses in Section IV use this set. The third part runs every model on the same sample of test articles, on CPU only (Section III-I).

Table II lists every data set of the study and its role. All choices of the StaticKD recipe were made on the development set, the development split of EMMediaTopic (1,000 articles in four languages with GPT-4o labels), and the recipe was fixed *before* the test set was built. No hyperparameter was selected on the test set. One further set informed two choices, the temperature $\tau = 1$ and the three-seed ensemble (Section III-F): 320 articles in 16 languages labeled by an LLM (Claude) with the prompt of [2]. The teacher was trained on labels that an LLM (GPT-4o) produced with the same prompt, so these labels are not independent of the teacher, and no reported result uses this set.

B. Problem Formulation

Given a news article d in any language, consisting of its title and body truncated to the first 512 words as in [2], a model predicts one label $y \in \mathcal{Y}$ among the 17 top-level IPTC topics. The teacher f_T outputs logits $\mathbf{z}^T(d) \in \mathbb{R}^{17}$. The goal is a student f_S whose inference costs $O(n)$ additions for n tokens, without matrix multiplication and without a deep learning

TABLE I
POSITION OF STATICKD RELATIVE TO RELATED APPROACHES

Approach	Model at inference time	Multilingual	Cross-lingual transfer	No deep learning library	CPU cost reported	IPTC labels
Kuzman & Ljubešić [2]	XLM-R-large (559.9M parameters)	yes	yes	no	no	yes
DistilBERT, MiniLMv2 [6], [7]	6-layer transformer	partly	yes	no	partly	no
Sparse Distillation [23]	deep averaging network + n-gram table	no (en)	no	yes	yes	no
Model2Vec [11], Wada et al. [12]	static embeddings, optional MLP classifier	yes	yes	yes	partly	no
fastText [8]	bag of n-grams	per language	no	yes	yes	no
StaticKD (proposed)	table of 17 logits per token	yes	yes	yes	yes	yes

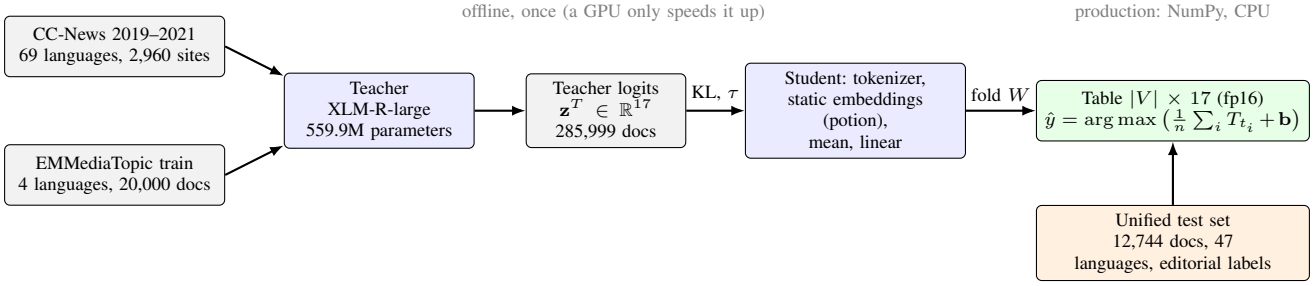


Fig. 1. StaticKD pipeline. The teacher labels an unlabeled news corpus once, offline. The student learns to imitate the teacher distribution, and its linear head is then folded into a token table, so inference only looks up and averages table rows. The final evaluation uses a test set whose labels do not come from the teacher.

TABLE II
DATA SETS OF THE STUDY AND THEIR ROLES

Set	Docs	Lang.	Labels	Used for
Distillation data	285,999	69	teacher	training the students
internal validation	8,579	–	teacher	choosing the epoch
Development (EMMediaTopic)	1,000	4	GPT-4o	choosing the recipe
CC-News held-out	12,554	63	teacher	agreement, unseen sites
Unified test set	12,744	47	editorial	all accuracy results
LLM check set	320	16	Claude	informal check only
English section sample	754	1	JSON-LD	label validity only

library, and whose accuracy against independent labels is as close as possible to that of f_T.

C. Distillation Data and Development Set

EMMediaTopic 1.0 [2], [9] contains Catalan, Greek, Croatian, and Slovenian articles labeled by GPT-4o. Its training split (20,000 articles) is part of the distillation data. Its development split (1,000 articles, median 218 words) was never used to train the teacher or the student and serves as the development set for choosing the recipe.

Most of the distillation data come from CC-News 2019–2021, a multilingual split [36] of the news archive of Common Crawl. The test set later draws on the same archive for 2024–2026 (CC-NEWS 2024–2026, Section III-D). CC-News 2019–2021 was processed with the following steps.

1) Keep only documents whose metadata language agrees with fastText language identification [8].
2) Take the title and body with at least 75 words (200 characters for ja, zh, th, my) and truncate to 512 words.
3) Keep at most 150 documents per site.
4) Form a held-out split by domain: all documents of some sites are set aside, at most 20% per language.
5) Add 156,157 documents from training sites only (at most 5,000 per language and 400 per site).

The distillation data contain 285,999 documents in 69 languages from 2,960 sites, with a median of 269 words. A few multilingual sites (for example bbc.com) were set aside in one language but also appear in another language of the distillation data, so their articles were removed from the held-out split. The resulting held-out set has 12,554 documents in 63 languages from 217 sites that never occur in the distillation data. This set has no independent labels, so it measures only *agreement* with the teacher, defined as the share of documents on which a model predicts the same top label as the teacher.

D. Unified Test Set Construction

The first source is CC-NEWS 2024–2026, sampled as one archive file (WARC) per month from January 2024 to August 2026 (32 files) of the Common Crawl CC-NEWS archive [37]. From each page, the structured metadata blocks in schema.org JSON-LD format [16] of type `NewsArticle` and its subtypes were extracted, with the fields `articleSection`, `BreadcrumbList`, `inLanguage`, `datePublished`, `isAccessibleForFree`, and `articleBody`. The text comes from `articleBody` when it has at least 30 words and is otherwise extracted with Trafilatura [38]. Sites that occur in the distillation data or in another evaluation set were removed, which leaves 399,468 articles from new sites.

The second source is public datasets. MasakhaNEWS [17] and MN-DS [18] are labeled by humans. L3Cube-IndicNews (LDC split) [19], Bangla news [20], and IndoNews [39] carry publisher categories.

Section and category labels record where the newsroom placed an article when it was published. Such labels are called *editorial labels* in this paper. No model that reads the article text produces these labels, so an evaluation on them is not circular with respect to the teacher. The only added step is translating section names onto the official IPTC Media Topic tree (1,085 active nodes). Section names are normalized (case,

diacritics) and mapped conservatively. A name is mapped only if the content usually found under that section belongs to one top-level topic. This gives 814 names mapped to 79 nodes. The table deliberately leaves 62 ambiguous names unmapped, for example *tech*, which can be *science and technology* or the information technology industry under *economy*. Likewise, *entertainment* can be *arts* or *lifestyle*, and *terrorism* can be *conflict* or *crime*. Astrology sections are rejected. An automatic check verifies that every target node exists in the IPTC tree, is not retired, and lies under the intended top-level topic. Articles whose sections map to different labels are rejected. MN-DS needs no mapping, because its labels already are IPTC topics. The mapping table was drafted with artificial intelligence assistance, and the first author reviewed all of its 916 rules (814 mapped names and 102 ambiguous, language-specific, and rejecting rules) and accepted them without changes.

Quality and leakage filters are applied in order.

1) Publication date on or after 1 January 2024 (CC-NEWS only).
2) Not behind a paywall.
3) Declared language agrees with fastText language identification when the language is supported.
4) At least 75 words.
5) Deduplication of uniform resource locators (URLs), identical texts, and near-duplicates (MinHash [31], Jaccard ≥ 0.7 [32]) across all sources.
6) Contamination: an article is dropped when $\geq 30\%$ of its word 8-grams occur in any of about 300 thousand training or evaluation documents. This filter removed 50 articles, mostly from MasakhaNEWS, which also draws on BBC.

Sampling takes at most 50 articles for each pair of language and label, preferring human-labeled sources, then publisher categories, then JSON-LD sections. Languages with fewer than 30 articles are dropped. The number of articles per site is not capped, because most language–label pairs come from few sites and such a cap would discard data without adding diversity. Articles from the same site tend to resemble each other, so the evaluation resamples whole sites when it computes confidence intervals (Section III-H), and Section IV-A repeats the comparison with at most five articles per site.

The final test set contains 12,744 articles in 47 languages, with 17 labels and 1,227 sites or sources and a median of 313 words. Of these, 4,094 articles (32.1%) carry human annotations, 2,600 (20.4%) publisher categories, and 6,050 (47.5%) JSON-LD sections. The type of label of an article (human annotation, publisher category, or JSON-LD section) is called its *label origin*. By the amount of distillation data in their language, 13 languages have none (for example ha, yo, ig, bn, pa), 9 have fewer than 500 documents, and 25 have at least 500. Site diversity is measured by the effective number of sites, $1/\sum_s q_s^2$, where q_s is the share of the articles of a language that come from site s. A source dataset without URLs counts as one site. By this measure, 13 languages have at least 10 effective sites, while 27 languages come from practically one publisher. Table III summarizes the sources and Fig. 2 shows how many articles each language has per label.

TABLE III
COMPOSITION OF THE UNIFIED TEST SET BY SOURCE

Source	Label origin	Articles	Languages	Sites/sources
CC-NEWS 2024–2026	JSON-LD section	6,050	29	1,092
MasakhaNEWS [17]	human annotation	3,339	16	6
L3Cube-IndicNews [19]	publisher category	2,361	10	1
MN-DS [18]	human annotation	755	1	127
Bangla news [20]	publisher category	139	1	1
IndoNews	publisher category	100	1	1
Total		12,744	47	1,227

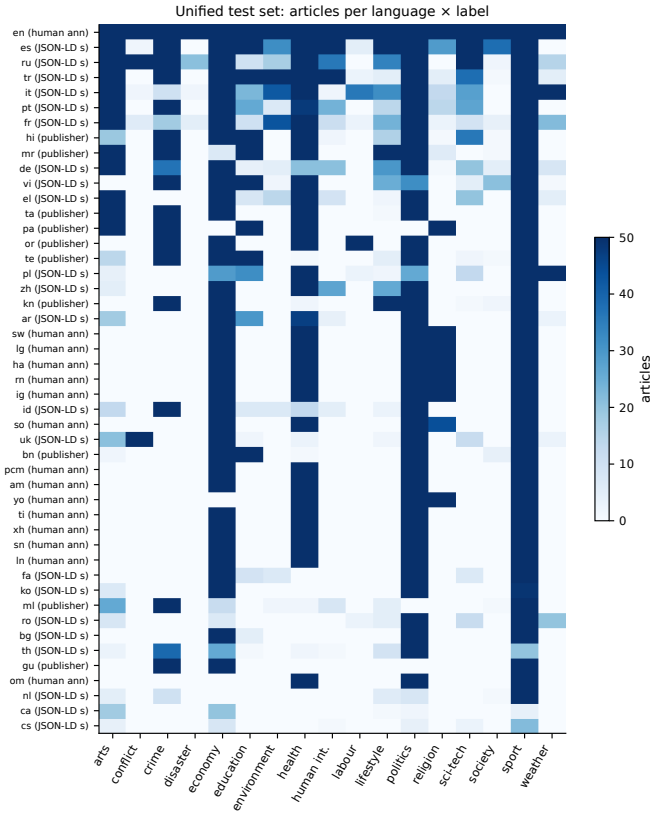


Fig. 2. Articles in the unified test set per language and label (at most 50 per language–label pair). The label origin in brackets is the dominant one for that language.

E. Teacher and Distillation Labels

The teacher is the IPTC classifier of [2], published as classla/multilingual-IPTC-news-topic-classifier. It is an XLM-RoBERTa-large model with 24 layers and 559.9 million parameters, and its architecture limits the input to 512 XLM-R tokens. The 512-word input of 60.7% of the test articles exceeds this limit, and for those articles the teacher reads about 287 of 438 words (medians). For every distillation document the teacher produces a logit vector $\mathbf{z}^T(d) \in \mathbb{R}^{17}$, which is stored in full. Labeling ran once in fp16 on an NVIDIA RTX 4050 Laptop GPU (6 GB), with batches sorted by text length. Measured on a fixed sample of 5,000 documents, the teacher labels 50.1 documents per second, so labeling the whole distillation data takes about 1.58 GPU hours. The teacher label distribution on CC-News 2019–2021 is plausible (*politics* 16.5%, *economy* 12.7%, *sport* 11.6%, *weather* 1.3%),

and its mean confidence per language ranges from 0.90 to 1.00.

F. The StaticKD Student

The student is built on potion-multilingual-128M, the multilingual static embedding model of Model2Vec [11], hereafter called potion. A document d is tokenized into $t_1, \ldots, t_n$ with the Unigram tokenizer of potion, which has 500,353 tokens, so that every token has a pre-trained embedding. Special tokens are not used, unknown tokens are dropped, and the length is capped at 1,024 tokens. Averaging has no position limit, so the cap only bounds the cost, and only 5.4% of the test articles exceed it. Each token has an embedding $\mathbf{e}_t \in \mathbb{R}^{256}$, and

$$\mathbf{h}(d) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{e}_{t_i}, \qquad \mathbf{z}^S(d) = W\,\mathrm{drop}(\mathbf{h}(d)) + \mathbf{b}, \quad (1)$$

with $W \in \mathbb{R}^{17\times 256}$ and dropout 0.1 during training. The embedding matrix E is initialized from potion, whose embeddings for 101 languages are distilled from BGE-M3 [13]. This initialization places the tokens of these 101 languages in one shared space before training starts. Section IV-C compares it with a random initialization.

The student learns from the stored teacher logits with the loss

$$\mathcal{L}_{\mathrm{KD}} = \tau^2\,\mathrm{KL}\big(p^T \,\big\|\, p^S\big), \qquad \tau = 1, \quad (2)$$

where $p^T = \mathrm{softmax}(\mathbf{z}^T/\tau)$ and $p^S = \mathrm{softmax}(\mathbf{z}^S/\tau)$ are the teacher and student distributions. The value $\tau = 1$ was chosen during development, and Section IV-C reports $\tau \in \{0.5, 2, 4\}$. E, W, and $\mathbf{b}$ are trained together. E uses SparseAdam ($\eta = 10^{-3}$), which updates only the rows of tokens that occur in the current batch. Tokens that never occur in the training documents therefore keep their potion embeddings, a property that Section V returns to. W and $\mathbf{b}$ use AdamW ($\eta = 3\cdot 10^{-3}$, weight decay 10^{-4}). The batch size is 64 and training runs for 6 epochs. As a cheap augmentation, each token of a training document is dropped with probability 0.2 at every step (token dropout), so that the model does not rely on a few keywords. Internal validation uses 3% of the distillation data (8,579 documents), and the epoch with the highest agreement with the teacher is kept.

Because the classification head is linear, (1) can be rewritten as

$$W\Big(\frac{1}{n}\sum_i \mathbf{e}_{t_i}\Big) + \mathbf{b} = \frac{1}{n}\sum_i T_{t_i} + \mathbf{b}, \quad (3)$$

with the table $T = EW^\top \in \mathbb{R}^{|V|\times 17}$. The folding step computes T once after training. Row T_t then holds the 17 topic scores that token t contributes to every article that contains it. The production model consists only of T (fp16), $\mathbf{b}$, and the tokenizer, and Algorithm 1 lists the whole inference procedure. The same argument covers an ensemble. Averaging the logits of K students trained with different seeds gives the same result as one student with table $\bar{T} = \frac{1}{K}\sum_k T^{(k)}$ and bias $\bar{\mathbf{b}}$, because (3) is linear in T. Three seeds (0, 1, and 2) were trained during development, and the final StaticKD model averages their tables ($K = 3$). It has 8.5 million parameters and a size of 34 MB.

Algorithm 1 StaticKD Inference for One Article

Require: article text d; tokenizer; table $\bar{T} \in \mathbb{R}^{|V|\times 17}$; bias $\bar{\mathbf{b}} \in \mathbb{R}^{17}$

1: $(t_1, \ldots, t_n) \leftarrow$ first 1,024 tokens of tokenize(d), without unknown tokens
2: $\mathbf{z} \leftarrow \frac{1}{n}\sum_{i=1}^{n} \bar{T}_{t_i} + \bar{\mathbf{b}}$ ▷ look up and average table rows
3: **return** $\arg\max_y\ z_y$ ▷ one of the 17 IPTC topics

TABLE IV
HYPERPARAMETERS OF ALL MODELS

Component	Value
StaticKD	
initial embedding	potion-multilingual-128M (500,353 × 256)
max. tokens / vector dropout	1,024 / 0.1
loss, τ	KL, $\tau = 1$
token dropout	0.2
optimizer embedding / head	SparseAdam 10^{-3} / AdamW $3\cdot 10^{-3}$, wd 10^{-4}
batch, epoch, validation	64, 6, 3% of data (selects the epoch)
ensemble	mean of 3 seed tables
fastText	
word tokens (fixed)	dim 100, lr 0.5, 15 epoch
subword tokens (fixed)	dim 100, lr 0.5, 15 epoch
GPT-4o labels	dim 100, lr 0.5, 25 epoch
word n-grams, loss	2, softmax, 2M buckets
tokens	words (+ character 2–5-grams) or XLM-R subwords
quantization	`quantize`, cutoff 300k, retrain
TF-IDF + SVM	
features	XLM-R subword uni+bigrams, min. df 3, max. 1,000,000, sublinear tf
classifier (fixed)	LinearSVC, $C = 0.50$
MiniLM-L6 / DistilmBERT + KD	
optimizer	AdamW $5\cdot 10^{-5}$, wd 0.01, warmup 6%, linear, fp16
batch, epoch, max. tokens	32, 2, 256
τ	$\{1, 2\}$, chosen on dev

The internal structure of the Unigram tokenizer needs much more random-access memory (RAM) than the table itself (16 MB). A compact variant shrinks that structure by removing token pieces that occur fewer than 10 times in the distillation data, keeping single characters. Its vocabulary has 335,441 tokens and the model takes 23 MB. Table IV lists the hyperparameters of all models.

G. Baselines

All baselines marked + KD are trained on the same distillation data (285,999 documents) with targets from the same teacher. fastText + KD [8] is trained on the top teacher label with word tokens (plus character n-grams) or XLM-R subwords, together with a quantized variant (`.ftz`). As a reference equivalent to the labeled data available to [2], fastText is also trained only on the GPT-4o labels of the EMMediaTopic training split. TF-IDF + SVM + KD is the non-neural baseline of Kuzman and Ljubešić (TF-IDF features with a support vector classifier), trained on the top teacher label over XLM-R subwords. The hyperparameters of these non-transformer baselines were set once, as listed in Table IV, without a search. MiniLM-L6 + KD uses mMiniLMv2-L6-H384, distilled from XLM-R-large [7], with the temperature $\tau \in \{1, 2\}$ chosen on the development set. DistilmBERT + KD uses `distilbert-base-multilingual-cased` [6] with the same recipe. Both transformers read at most 256 tokens, which truncates 91.1% (MiniLM-L6) and 93.2% (DistilmBERT) of the test articles to about 142 and 116 words. Both are exported to ONNX fp32 for the CPU benchmark. The ONNX int8 teacher is a dynamic int8 quantized export of the teacher and is its fastest CPU configuration.

Sparse Distillation [23], the closest prior work, and the classifier trainer of Model2Vec were not run as separate baselines (Section VI). Parts of both are still covered. fastText + KD is a bag-of-n-grams model with averaging and a linear head, trained on the same teacher labels, and the random-initialization ablation (Section III-J) trains the embedding table from scratch, as Sparse Distillation does. The frozen-embedding ablation trains only a linear head on top of potion and thus acts as an off-the-shelf Model2Vec baseline with a distilled head.

H. Evaluation Protocol and Statistical Tests

The main metric is macro-F1 over the labels present in the reference, as in the evaluation code of [2], because the label distribution is imbalanced. The evaluation also reports accuracy, macro-F1 per language averaged over languages so that large languages do not dominate, and agreement with the teacher on the held-out set.

95% confidence intervals come from a percentile bootstrap with 1,000 resamples. On the test set, the resampling unit is the site, or the source dataset when no URL is available. This site-cluster bootstrap [35] draws whole sites, so related articles from one site stay together and the intervals are not made too narrow by treating them as independent. On the development set, the unit is the document. Two models on the same documents are compared with a paired (cluster) bootstrap with 2,000 resamples for Δmacro-F1 [33] and an exact McNemar test for accuracy [40], with Holm correction [41] within each set. Each ablation variant uses five seeds, and the final recipe uses ten. Differences between variants are tested with Welch tests at the seed level, Holm-corrected. Whether a difference holds consistently across languages is tested with a Wilcoxon signed-rank test on per-language macro-F1 and accuracy [34]. Subgroup analyses cover label origin, region, the amount of distillation data of the language, teacher pre-training coverage (whether the language is among the 100 languages on which XLM-R was pre-trained [3]), site diversity, script, language, and label. Per-language properties are related to performance with Spearman correlations and, across scripts, with a Kruskal–Wallis test. Two per-language quantities are used there. The *accuracy gap* is the StaticKD accuracy minus the teacher accuracy, so a negative gap means that StaticKD is less accurate. The *token coverage* is the share of token occurrences in the test articles of a language whose token occurs at least 10 times in the distillation data; the embeddings of rarer tokens are rarely updated during training (Section III-F). Calibration is measured with the expected calibration error (ECE, 15 bins) [42], because the cascade mode (Section III-I) routes articles by the confidence of StaticKD.

I. Efficiency Protocol

The benchmark runs on CPU only, with the GPU hidden. Each model runs in a fresh process, and the thread count is pinned in every library (OMP, MKL, OpenBLAS, ONNX Runtime, `tokenizers`). Latency is measured with one article per call, as in a streaming setting (p50 and p95), and throughput with batches of 64, or 8 for transformers. Model RAM is the increase in resident set size after loading the model. The text sample is drawn at random from the unified test set and is the same for every model: 30 latency and 48 throughput documents for the teacher and the transformers, and 500 and 3,000 for the light models. All models are measured in one session on an idle machine. StaticKD reads up to 1,024 tokens per article while the teacher stops at 512, so the speed comparison does not favor StaticKD. In cascade mode, StaticKD decides the articles on which its top probability is at least a threshold θ and passes the rest to the ONNX int8 teacher. With a share s of articles decided by StaticKD and single-core throughputs r_S of StaticKD and r_T of the teacher, the effective throughput of the cascade is $1/(1/r_S + (1-s)/r_T)$.

The machine is a laptop with an Intel Core Ultra 7 155H, 31 GB of RAM, and Windows 11. This CPU is hybrid, with 6 performance cores (P-cores), 8 slower efficiency cores (E-cores), and 2 low-power E-cores. Without pinning, the operating system moves threads to the slow cores. Each process is therefore pinned to physical P-cores (logical CPU 0 for one core; logical CPUs 0, 10, 12, and 14 for four cores), which approximates a small server with uniform cores.

J. Experimental Design and Implementation Details

Table IX lists the 60 runs of the final recipe and its ablations. Each ablation variant changes one factor of the final recipe: hard labels instead of soft labels; EMMediaTopic data only (4 languages, 10 epochs); 130 thousand documents without the extra CC-News 2019–2021 data; random initialization with the same standard deviation as potion; frozen embeddings; no token dropout; a 512-unit multilayer perceptron (MLP) head; and $\tau \in \{0.5, 2, 4\}$. Twelve further runs during development tested four additions to the recipe with three seeds each (bigram features, idf-weighted pooling, documents weighted by teacher confidence, and an extra loss on the GPT-4o labels). None of them differs significantly from the final recipe on the development set, and none changes the test macro-F1 by more than 0.004, so these variants are left out of the recipe and of Table IX. One StaticKD run takes 5.2 minutes on average on the same GPU. The implementation uses Python 3.12, NumPy 2.5, PyTorch 2.6, transformers 5.17, `tokenizers` 0.23, ONNX Runtime 1.30, scikit-learn 1.9, and fastText 0.9.2. The seed, library versions, and training history of every run are stored with the model.

IV. Results

A. Accuracy of All Models (RQ1)

StaticKD keeps 91.1% of the teacher macro-F1 on the unified test set, with 0.545 [0.490–0.605] against 0.598 [0.546–0.643] (Table V). Both scores lie far below the development scores (0.761 and 0.822), because most of the test set lies outside the languages, sources, and period of the teacher training data. On the development set StaticKD keeps 92.6% of the teacher score, and on the 63 held-out languages it agrees with the teacher on 81.1% of the documents. A single student

TABLE V
MAIN RESULTS. MACRO-F1 [95% CI] AND ACCURACY ON THE DEVELOPMENT SET (EMMEDIATOPIC DEV, GPT-4O LABELS), AGREEMENT WITH THE TEACHER ON CC-NEWS HELD-OUT, AND RESULTS ON THE UNIFIED TEST SET (47 LANGUAGES, INDEPENDENT LABELS). TEST CIS FROM THE SITE-CLUSTER BOOTSTRAP

	dev (4 languages)		held-out	unified test (47 languages)			
Model	Macro-F1 [95% CI]	Accuracy	Agreement	Macro-F1 [95% CI]	Accuracy	Macro-F1/language	% teacher
Teacher XLM-R-large	0.822 [0.779–0.851]	0.859	–	0.598 [0.546–0.643]	0.700	0.705	100.0%
Teacher XLM-R-large, ONNX int8	0.823 [0.781–0.852]	0.858	–	0.600 [0.547–0.646]	0.699	0.702	100.3%
DistilmBERT + KD	0.753 [0.710–0.785]	0.801	0.799	0.534 [0.481–0.618]	0.623	0.602	89.4%
MiniLM-L6 + KD	0.785 [0.737–0.815]	0.820	0.825	0.566 [0.511–0.624]	0.667	0.661	94.6%
StaticKD (proposed)	**0.761 [0.717–0.792]**	0.800	0.811	**0.545 [0.490–0.605]**	0.641	0.622	91.1%
StaticKD-compact	0.754 [0.710–0.786]	0.797	0.802	0.484 [0.421–0.597]	0.544	0.507	81.0%
fastText + KD, subword, quantized	0.708 [0.664–0.739]	0.759	0.740	0.435 [0.370–0.556]	0.512	0.464	72.7%
fastText + KD, subword	0.727 [0.691–0.756]	0.769	0.751	0.445 [0.384–0.563]	0.522	0.473	74.4%
fastText + KD, word	0.743 [0.707–0.773]	0.789	0.753	0.438 [0.373–0.560]	0.511	0.472	73.3%
TF-IDF + SVM + KD	0.755 [0.719–0.786]	0.798	0.776	0.486 [0.431–0.585]	0.567	0.522	81.3%
fastText, GPT-4o labels, subword	0.665 [0.613–0.700]	0.729	0.260	0.086 [0.069–0.119]	0.150	0.118	14.4%
fastText, GPT-4o labels, word	0.651 [0.601–0.693]	0.738	0.307	0.106 [0.085–0.150]	0.181	0.142	17.7%

without ensembling reaches 0.542 $\pm$ 0.004 on the test set (10 seeds).

MiniLM-L6 (0.566), which starts from a distillation of XLM-R-large, lies between StaticKD and the teacher. DistilmBERT (0.534) is on par with StaticKD while being much more expensive. The two baselines of similar cost, fastText + KD and TF-IDF + SVM + KD, fall below StaticKD. The last two rows show what happens without teacher labels in other languages. fastText trained only on the GPT-4o labels of four languages reaches 0.665 on the development set but only 0.086 on the test set.

The paired tests in Table VI confirm this ranking. Against the teacher, Δmacro-F1 is -0.053 [-0.065, -0.033], and against MiniLM-L6 it is -0.021 [-0.038, -0.008] ($p = 0.004$). The difference from DistilmBERT (+0.010, $p = 0.800$) is not significant. StaticKD is correct on more test articles than DistilmBERT (McNemar $p < 0.001$), but the McNemar test treats articles as independent and ignores the dependence between articles of the same site. StaticKD is better than the quantized fastText + KD by +0.110 [+0.042, +0.151] and than TF-IDF + SVM by +0.059 [+0.009, +0.093] ($p = 0.030$).

The advantage over fastText + KD holds in almost every language, since StaticKD has the higher macro-F1 in 43 of 47 languages (Wilcoxon $p < 0.001$). The advantage over TF-IDF + SVM is narrower. StaticKD has the higher macro-F1 in 31 of 47 languages (Wilcoxon $p = 0.003$), but per-language accuracy is higher in 24 languages and lower in 22 (Wilcoxon $p = 0.127$), so the two models are close on accuracy. Against DistilmBERT neither metric differs consistently across languages (Wilcoxon $p = 0.604$ for macro-F1 and 0.322 for accuracy). StaticKD beats the teacher in only 7 of 47 languages.

The share of the teacher score that StaticKD keeps varies far more with the amount of distillation data in the language of the article than with the type of label (Table VII, Fig. 3). By label origin it ranges from 89.5% (publisher categories) to 95.8% (JSON-LD sections). By the amount of distillation data in the language, it falls from 97.5% in languages with at least 500 documents to 72.3% in languages with none. Limiting the test set to at most five articles per site and language–label pair (8,139 articles) leaves the model ranking unchanged (StaticKD 0.587, teacher 0.627).

Table VIII splits the test set by source. On MN-DS, the only source annotated directly with IPTC topics, StaticKD keeps 97.4% of the teacher score (0.535 against 0.549), and the teacher itself reaches its lowest macro-F1 of all sources there. StaticKD keeps the smallest share on MasakhaNEWS (80.0%). Its human labels follow a non-IPTC scheme, and 10 of its 16 languages are absent from the distillation data. Bangla is absent as well, and Bangla news shows most clearly what the pre-trained embeddings contribute. There, fastText + KD and TF-IDF + SVM collapse to 0.150 and 0.010, while StaticKD reaches 0.748.

B. Validity of the Editorial Labels

Two thirds of the test labels are section and category labels, which record newsroom decisions rather than annotation under a labeling guideline. This subsection checks whether such labels behave like human labels in the same language. The check compares how well the same teacher agrees with each label type, so the teacher acts only as a fixed reference. If section labels were much noisier than human labels, the teacher would agree with them clearly less often than with human labels in the same language.

Because the sampling prefers human-labeled sources (Section III-D), the English part of the test set contains only human labels. For a comparison within English, a separate sample of 754 English CC-NEWS articles with JSON-LD section labels was therefore drawn from 306 sites outside the test set, at most 50 per label. The teacher reaches an accuracy of 0.646 on the section labels (macro-F1 0.617) and 0.559 on the MN-DS and MasakhaNEWS human labels (macro-F1 0.565). Accuracy on section labels is higher for 12 of the 17 labels (Wilcoxon $p = 0.033$), and the ranking of label difficulty correlates between the two label types ($\rho = 0.63$, $p = 0.007$). The hard labels, *society*, *human interest*, and *religion*, are hard

TABLE VI
PAIRED TESTS OF STATICKD AGAINST OTHER MODELS ON THE UNIFIED TEST SET (Δ = STATICKD − COMPARISON MODEL; p HOLM-CORRECTED OVER THE COMPARISONS; b/c = DOCUMENTS CORRECT ONLY FOR STATICKD / ONLY FOR THE COMPARISON MODEL)

StaticKD vs	Δmacro-F1 [95% CI]	p bootstrap	Δaccuracy	McNemar b/c	p McNemar
Teacher XLM-R-large	−0.053 [−0.065, −0.033]	<0.001	−0.059	654/1400	<0.001
Teacher XLM-R-large, ONNX int8	−0.055 [−0.067, −0.035]	<0.001	−0.058	644/1386	<0.001
DistilmBERT + KD	+0.010 [−0.026, +0.041]	0.800	+0.018	1372/1143	<0.001
MiniLM-L6 + KD	−0.021 [−0.038, −0.008]	0.004	−0.026	866/1197	<0.001
StaticKD-compact	+0.060 [+0.003, +0.100]	0.030	+0.097	1345/110	<0.001
fastText + KD, subword, quantized	+0.110 [+0.042, +0.151]	<0.001	+0.129	2157/519	<0.001
fastText + KD, subword	+0.100 [+0.037, +0.138]	<0.001	+0.119	2021/505	<0.001
fastText + KD, word	+0.106 [+0.041, +0.149]	<0.001	+0.130	2176/513	<0.001
TF-IDF + SVM + KD	+0.059 [+0.009, +0.093]	0.030	+0.074	1667/719	<0.001
fastText, GPT-4o labels, subword	+0.458 [+0.415, +0.496]	<0.001	+0.491	6597/344	<0.001
fastText, GPT-4o labels, word	+0.439 [+0.393, +0.475]	<0.001	+0.460	6252/391	<0.001

TABLE VII
TEST MACRO-F1 BY SUBGROUP. LAST COLUMN: STATICKD MACRO-F1 AS A PERCENTAGE OF THE TEACHER MACRO-F1 IN THE SAME SUBGROUP

Subgroup	n	Teacher	DistilmBERT	MiniLM	**StaticKD**	fastText (q)	TF-IDF+SVM	fastText GPT	% teacher
Label origin									
human annotation	4,094	0.463	0.436	0.436	0.414	0.336	0.393	0.027	90%
publisher category	2,600	0.702	0.498	0.674	0.629	0.319	0.330	0.063	90%
JSON-LD section	6,050	0.651	0.629	0.634	0.624	0.579	0.609	0.136	96%
Distillation data of the language									
sufficient (≥500)	7,644	0.633	0.617	0.616	0.618	0.572	0.593	0.116	98%
few (<500)	2,243	0.537	0.344	0.496	0.478	0.202	0.227	0.037	89%
none (0)	2,857	0.557	0.452	0.502	0.402	0.174	0.230	0.048	72%
Region									
Africa	3,044	0.697	0.532	0.576	0.542	0.312	0.408	0.085	78%
South Asia	2,681	0.575	0.410	0.544	0.509	0.246	0.276	0.046	88%
East/Southeast Asia & Middle East	1,574	0.636	0.620	0.621	0.622	0.560	0.629	0.085	98%
Europe	5,445	0.653	0.637	0.638	0.638	0.595	0.610	0.136	98%
Teacher pre-training coverage									
outside XLM-R pre-training	1,750	0.719	0.586	0.620	0.501	0.317	0.457	0.086	70%
in XLM-R pre-training	10,994	0.609	0.549	0.585	0.576	0.468	0.503	0.096	95%
Site diversity									
diverse (≥10)	5,225	0.657	0.642	0.643	0.642	0.598	0.616	0.134	98%
single publisher (<3)	6,204	0.517	0.387	0.475	0.433	0.221	0.287	0.045	84%
medium (3–10)	1,315	0.559	0.556	0.539	0.543	0.503	0.533	0.089	97%

TABLE VIII
TEST MACRO-F1 BY SOURCE OF THE UNIFIED TEST SET. MN-DS IS THE ONLY SOURCE ANNOTATED DIRECTLY WITH IPTC TOPICS; MASAKHANEWS IS HUMAN-ANNOTATED UNDER ITS OWN SCHEME. LAST COLUMN: STATICKD AS A PERCENTAGE OF THE TEACHER

Source	Label origin	n	Lang.	Teacher	DistilmBERT	MiniLM	**StaticKD**	fastText (q)	TF-IDF+SVM	% teacher
CC-NEWS 2024–2026	JSON-LD section	6,050	29	0.651	0.629	0.634	0.624	0.579	0.609	96%
MasakhaNEWS [17]	human annotation	3,339	16	0.695	0.545	0.582	0.556	0.341	0.434	80%
L3Cube-IndicNews [19]	publisher category	2,361	10	0.709	0.501	0.680	0.639	0.328	0.342	90%
MN-DS [18]	human annotation	755	1	0.549	0.546	0.535	0.535	0.493	0.461	97%
Bangla news [20]	publisher category	139	1	0.835	0.730	0.817	0.748	0.150	0.010	90%
IndoNews	publisher category	100	1	0.885	0.846	0.856	0.862	0.852	0.873	97%

under both. By this measure, section labels are not noisier than human labels in English.

French offers a second comparison inside the test set. The teacher reaches an accuracy of 0.716 on 208 articles with section labels and 0.500 on 200 human-labeled MasakhaNEWS articles, the same direction as in English.

A last check looks at each source category, restricted to languages covered by XLM-R pre-training, to see why agreement is lower on the human-labeled sources. The median agreement of the teacher with the mapped label per category is 0.46 for MasakhaNEWS, 0.50 for MN-DS, 0.81 for L3Cube, and 0.87 for CC-NEWS sections. The lowest agreement occurs for source categories that are broader than their IPTC target. For example, only 45% of MasakhaNEWS *politics* articles are predicted as *politics*, with conflict (29%) as the main alternative. Agreement is also low for labels whose boundaries are unclear even to human annotators, such as *society* in MN-DS, in line with the low inter-annotator agreement reported for that label [2]. Such mismatches reflect differences between the category scheme of a source and IPTC. Since every model is scored against the same labels, comparisons between models are unaffected, and absolute scores read as agreement with the editorial categorization.

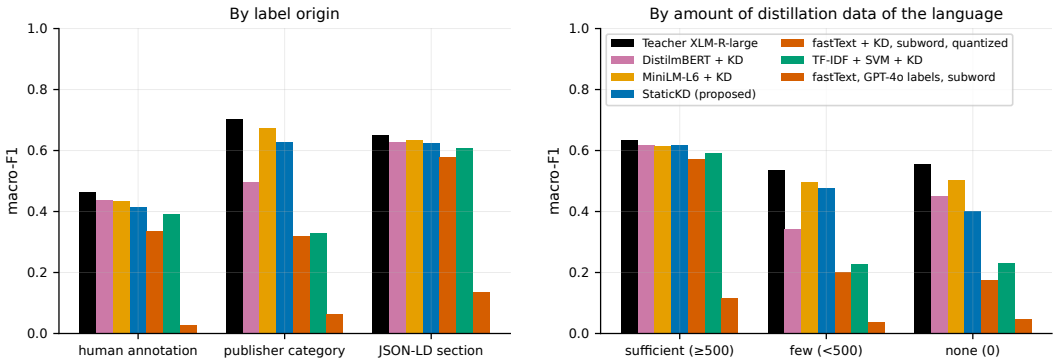


Fig. 3. Test macro-F1 by label origin (left) and by the amount of distillation data in the language of the article (right).

C. Ablation, Temperature, and Ensembling (RQ2)

Two components carry most of the accuracy: multilingual distillation data and the pre-trained embeddings (Table IX, five seeds per variant). Distilling on EMMediaTopic alone (4 languages) lowers the test macro-F1 by 0.051 and the held-out agreement by 0.077. Starting from random embeddings instead of potion lowers the test macro-F1 by 0.120 ($p < 0.001$), the largest drop of all variants.

Freezing the embeddings, so that only the linear head on top of potion is trained, gives a mixed result. It lowers the development score by 0.035 ($p < 0.001$) and the held-out agreement by 0.020 ($p < 0.001$), but raises the test macro-F1 by +0.014 ($p < 0.001$). Table X locates this gain in languages with little distillation data (+0.059) and with none (+0.037). In languages with at least 500 documents the frozen embeddings are slightly weaker (-0.007). The final recipe was fixed on the development set before this effect was observed, and Section V explains why that set cannot reveal it.

None of the remaining changes moves the test macro-F1 significantly. Hard teacher labels change the test macro-F1 by +0.005 ($p = 0.433$) and the development score by -0.006 ($p = 1.000$), so hard labels work as well as soft labels. Reducing the data to 130 thousand documents changes the test macro-F1 by -0.002 ($p = 1.000$), removing token dropout by +0.001 ($p = 1.000$), and an MLP head that cannot be folded by +0.009 ($p = 0.281$).

A higher temperature does not help (Fig. 4). Compared with $\tau = 1$, the common choice $\tau = 2$ [10] lowers the held-out agreement by 0.007 ($p < 0.001$), and its changes on the test macro-F1 (-0.005, $p = 0.059$) and on the development set (-0.006, $p = 0.705$) are not significant. With $\tau = 4$ all three metrics drop, the test macro-F1 by 0.014 ($p < 0.001$).

Averaged over seed combinations, growing the ensemble from one to three seeds moves the test macro-F1 from 0.542 to 0.544 and the held-out agreement from 0.808 to 0.810 (Fig. 5), and ten seeds give 0.546. The gain is small, but it costs nothing at inference time because every ensemble size is still one table.

D. CPU Efficiency (RQ3)

The teacher is too slow for a large stream on CPU, and StaticKD removes this cost almost entirely (Table XI, Figs. 6 and 7). In PyTorch fp32 the teacher needs 2,971 ms per article on one core, with a throughput of 0.33 articles per second. Its best CPU configuration, ONNX int8, needs 1,239 ms with a throughput of 0.77 articles per second. Classifying one million articles per day would take 15.1 cores (ONNX int8) to 34.8 cores (PyTorch) for this step alone.

StaticKD needs 2.03 ms per article and processes 473 articles per second per core. This is about 616 times faster than the ONNX int8 teacher, 1,420 times faster than the PyTorch teacher, and 29 times faster than MiniLM-L6. Its file is 16 times smaller than the ONNX int8 teacher, and one million articles per day need 0.02 cores. Its model RAM is 501 MB, mostly for the tokenizer structure. The compact variant reduces it to 277 MB. With one thread, tokenization (579 tokens per article on average) takes 98% of the prediction time, while averaging the table rows alone runs at 28,755 articles per second. The speed of StaticKD is thus limited by the tokenizer, not by the model.

StaticKD is not the fastest model in the table. The quantized fastText + KD is slightly faster (532 articles per second), smaller (18 MB), and uses less RAM (300 MB), but its test macro-F1 is 0.435 against 0.545 for StaticKD. Of the four models with comparable latency (StaticKD, its compact variant, fastText, and TF-IDF + SVM), StaticKD is the most accurate (Fig. 6).

E. Analysis by Language and Label (RQ4)

In a typical language StaticKD is close to the teacher, with a median per-language accuracy gap of -0.024 (Fig. 8). Table XII relates the gap to properties of the language. On the held-out set (53 languages with at least 20 documents), the agreement of StaticKD with the teacher rises with the log amount of distillation data ($\rho = 0.31$, $p = 0.022$) and with teacher confidence ($\rho = 0.71$, $p < 0.001$). It does not differ significantly across scripts (Kruskal–Wallis $H = 10.37$, $p = 0.065$). The weak spot is a group of African languages. On languages outside XLM-R pre-training that are also absent from the distillation data (ig, lg, ln, pcm, rn, sn, ti, yo), StaticKD reaches a mean accuracy of only 0.407, while the teacher reaches 0.643.

Four Indian languages (bn, ml, or, pa) show how StaticKD handles text whose tokens it has hardly seen in training. In these languages at least 68% of the token occurrences

TABLE IX
ABLATIONS AND TEMPERATURE (ONE STUDENT, NO ENSEMBLE). MEAN ± STANDARD DEVIATION OVER SEEDS; *: DIFFERENT FROM THE FINAL RECIPE (WELCH, HOLM, $p < 0.05$)

Variant	Seeds	Dev macro-F1	Held-out agreement	Test macro-F1	Test accuracy
Final recipe					
Final recipe (one seed)	10	0.755±0.010	0.808±0.001	0.542±0.004	0.640
Component ablation					
Hard teacher labels (cross-entropy)	5	0.749±0.007	0.805±0.001	0.547±0.004	0.643
EMMediaTopic only (4 languages, 20k)	5	0.723±0.006*	0.730±0.001*	0.491±0.002*	0.583
130k documents (no extra CC-News)	5	0.736±0.009*	0.799±0.002*	0.541±0.002	0.638
Randomly initialized embedding	5	0.715±0.005*	0.739±0.004*	0.422±0.006*	0.493
Frozen embedding (linear head only)	5	0.720±0.001*	0.788±0.001*	0.557±0.002*	0.661
No token dropout	5	0.757±0.004	0.808±0.001	0.544±0.002	0.640
512-unit MLP head (cannot be folded)	5	0.762±0.009	0.813±0.002*	0.551±0.006	0.649
Temperature					
$\tau = 0.5$	5	0.750±0.004	0.807±0.002	0.546±0.005	0.642
$\tau = 2$	5	0.748±0.003	0.801±0.001*	0.537±0.001	0.638
$\tau = 4$	5	0.723±0.004*	0.791±0.001*	0.529±0.002*	0.635

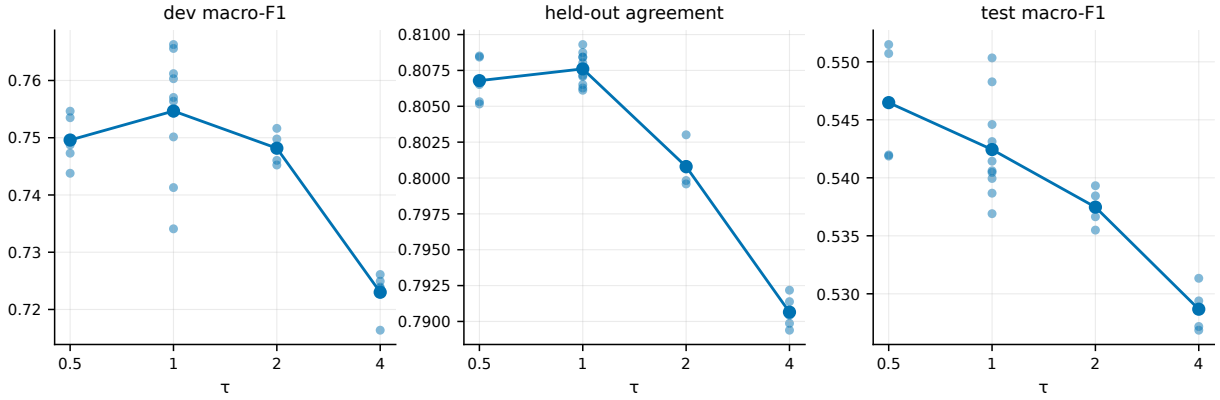


Fig. 4. Effect of the distillation temperature. Each dot is one seed; the line connects the means.

TABLE X
MACRO-F1 OF SELECTED VARIANTS BY THE AMOUNT OF DISTILLATION DATA IN THE LANGUAGE OF THE ARTICLE (TEST SET; MEAN OVER SEEDS)

Variant	Dev	Sufficient (≥500)	Few (<500)	None (0)
Final recipe (one seed)	0.755	0.615	0.482	0.403
Frozen embedding (linear head only)	0.720	0.607	0.541	0.440
Randomly initialized embedding	0.715	0.567	0.182	0.161
EMMediaTopic only (4 languages, 20k)	0.723	0.559	0.466	0.351
Hard teacher labels (cross-entropy)	0.749	0.617	0.482	0.410
512-unit MLP head (cannot be folded)	0.762	0.613	0.484	0.421
$\tau = 2$	0.748	0.612	0.476	0.363

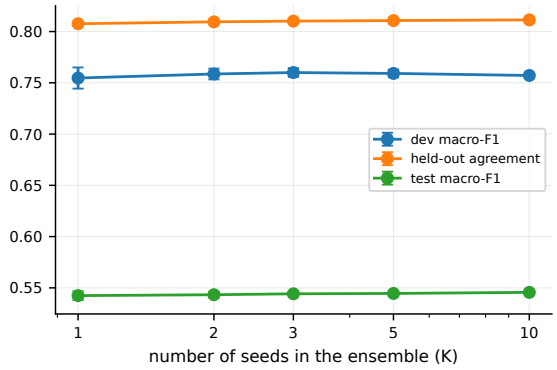


Fig. 5. Effect of the table ensemble size (mean ± standard deviation over seed combinations).

come from tokens seen fewer than 10 times in the distillation data. StaticKD still reaches a mean accuracy of 0.624 in these languages (teacher 0.738). The compact variant removes these rare tokens from the tokenizer, and its accuracy in these languages falls to 0.013. In the remaining languages the compact variant reaches 0.578 against 0.640 for the full StaticKD.

Per label (Fig. 9), the hardest labels for StaticKD are *society* (0.16), *disaster* (0.29), *human interest* (0.32), and *conflict* (0.37), and the easiest are *sport* (0.88), *weather* (0.84), and *education* (0.75). The largest gaps to the teacher occur for *environment* (−0.17), *religion* (−0.15), and *arts* (−0.08), and the most frequent confusions are *society* → *human interest*, *human interest* → *arts*, and *society* → *politics* (20% each).

The table also shows which tokens carry the topic signal of an article. The specificity of a token is defined as the largest difference between one entry of its table row and the mean of the other 16 entries. Keeping only the 10% most specific tokens of each article (about 58 tokens per article) leaves the test macro-F1 almost unchanged (0.543 against 0.545), whereas a random 10% of the tokens gives 0.435.

TABLE XI
CPU EFFICIENCY (PINNED P-CORES; SAMPLE FROM THE UNIFIED TEST SET; 1C/4C = 1/4 CORES)

Model	Latency p50 / p95, 1c (ms)	Throughput 1c (doc/s)	Throughput 4c (doc/s)	Model RAM (MB)	File (MB)	Dependencies
Teacher XLM-R-large	2,971 / 3,210	0.33	0.76	2,957	2,136	torch, transformers
Teacher XLM-R-large, ONNX int8	1,239 / 1,279	0.77	2.62	1,383	537	onnxruntime, transformers
DistilmBERT + KD	198 / 215	4.88	14.6	1,068	516	onnxruntime, transformers
MiniLM-L6 + KD	57.8 / 68.5	16.4	36.5	1,242	408	onnxruntime, transformers
StaticKD	**2.03 / 3.46**	**473**	**893**	501	**34**	numpy, tokenizers
StaticKD-compact	1.87 / 3.32	549	1,851	277	23	numpy, tokenizers
fastText + KD, subword, quantized	2.08 / 3.95	532	892	300	18	fasttext, tokenizers
fastText + KD, subword	2.26 / 4.30	455	592	1,121	839	fasttext, tokenizers
TF-IDF + SVM + KD	2.93 / 4.48	421	595	826	166	scikit-learn, tokenizers

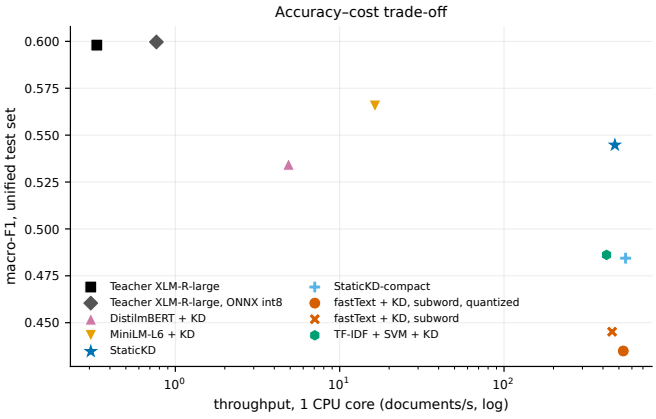


Fig. 6. Test macro-F1 on the unified test set against single-core CPU throughput (log scale).

TABLE XII
FACTORS BEHIND PER-LANGUAGE PERFORMANCE

Test	Statistic	p
Spearman: agreement vs log(distillation data)	$\rho = 0.31$	0.022
Spearman: agreement vs number of training sites	$\rho = 0.29$	0.033
Spearman: agreement vs teacher confidence	$\rho = 0.71$	<0.001
Kruskal–Wallis: across scripts	$H = 10.37$	0.065
Spearman (test): accuracy gap vs log(data)	$\rho = 0.52$	<0.001
Spearman (test): StaticKD accuracy vs token coverage	$\rho = 0.50$	<0.001
Spearman (test): accuracy gap vs token coverage	$\rho = 0.49$	<0.001

TABLE XIII
CASCADE STATICKD → ONNX INT8 TEACHER ON THE TEST SET (EFFECTIVE SINGLE-CORE THROUGHPUT)

θ	StaticKD share	Accuracy on share	Macro-F1	Accuracy	doc/s
0 (StaticKD only)	100.0%	0.641	0.545	0.641	473
0.5	80.8%	0.732	0.586	0.690	3.97
0.6	74.0%	0.760	0.593	0.697	2.93
0.7	67.4%	0.788	0.600	0.702	2.34
0.8	60.3%	0.813	0.602	0.703	1.92
0.9	51.7%	0.839	0.602	0.703	1.58
Teacher only	0.0%	–	0.600	0.699	0.77

F. Calibration and Cascade Mode

The ECE of StaticKD is 0.041 on the development set and 0.139 on the test set, lower than that of the teacher (0.090 and 0.245). Both models become overconfident on the test set, which lies outside the training distribution (Fig. 10). In cascade mode, StaticKD decides the articles on which its confidence is at least θ and sends the rest to the teacher (Table XIII). With $\theta = 0.8$, StaticKD decides 60.3% of the articles with an accuracy of 0.813. The cascade reaches an overall accuracy of 0.703 and a macro-F1 of 0.602, against 0.699 and 0.600 for the teacher alone, at an effective throughput of 1.92 articles per second per core against 0.77.

V. DISCUSSION

A table lookup keeps about nine tenths of the teacher macro-F1 at an inference cost two to three orders of magnitude lower (Sections IV-A and IV-D). This agrees with the finding that order-free representations are already competitive for topic classification [14]. The token analysis in Section IV-E shows why a table suffices: a small share of topic-specific tokens carries almost all of the topic signal of an article. The remaining gap to the teacher has two sources. The first is a few labels, *environment*, *religion*, and *arts*, on which StaticKD loses the most against the teacher (Fig. 9). The labels on which StaticKD scores lowest, *society*, *disaster*, *human interest*, and *conflict*, are a separate matter. They are also among the hardest labels for the teacher, and *society* and *human interest* are known to be difficult for human annotators as well [2], so their low scores are not specific to the student. The second source of the gap is languages that are underrepresented in the distillation data (Table VII).

With the same distillation data, MiniLM-L6 closes part of this gap, whereas DistilmBERT does not (Table VI). Both cost one to two orders of magnitude more than StaticKD. MiniLM comes from a distillation of XLM-R-large and DistilmBERT from mBERT, which suggests that the quality of the initial multilingual representation matters more than the number of layers.

The ablations show where the knowledge of the student comes from (RQ2). The largest source is the pre-trained multilingual initialization, whose removal lowers the test macro-F1 by 0.120, against 0.051 for the removal of the multilingual data. The multilingual distillation data labeled by the teacher come next, consistent with Sparse Distillation [23], which also relies on a large set of teacher-labeled text. The compact variant gives the most direct evidence for the role of the initialization. In Indian languages that hardly occur in the distillation data (for example bn, pa, or), most token occurrences come from tokens seen fewer than 10 times in the distillation data, so their embeddings are rarely or never updated and stay close to their potion values. StaticKD still classifies these languages with a mean accuracy of 0.624, yet the accuracy falls to 0.013 once the compact tokenizer removes those tokens. Transfer

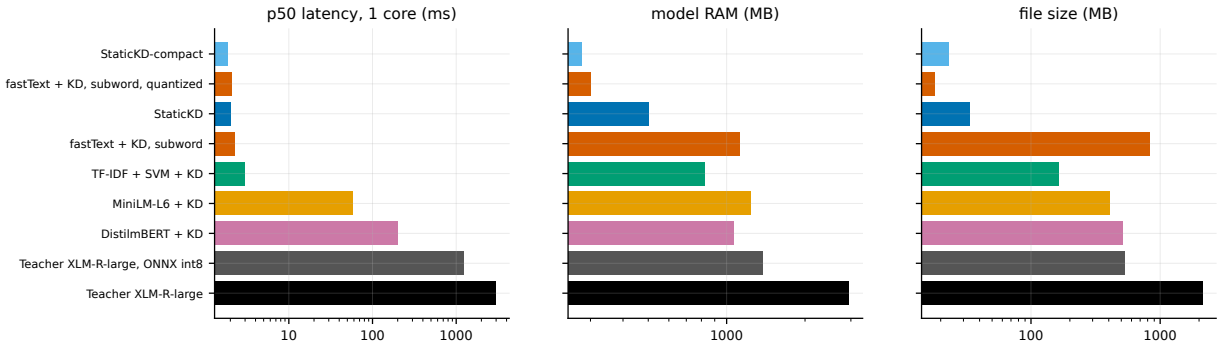


Fig. 7. Latency per article, model RAM, and file size (log scale).

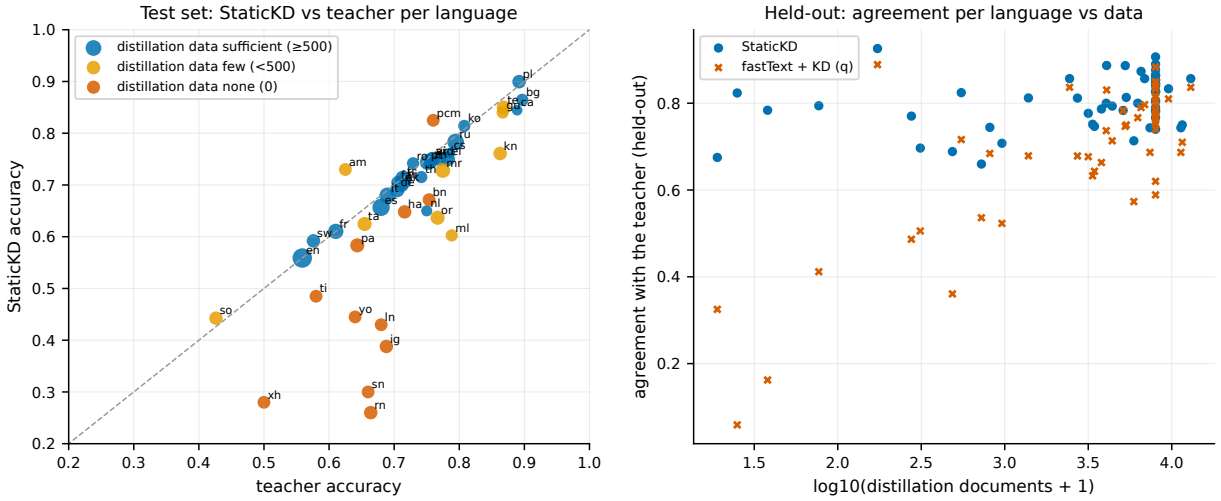


Fig. 8. Left: StaticKD accuracy against teacher accuracy per test language (color shows the distillation data of the language, size the number of articles). Right: held-out agreement per language against the amount of distillation data for StaticKD and fastText + KD.

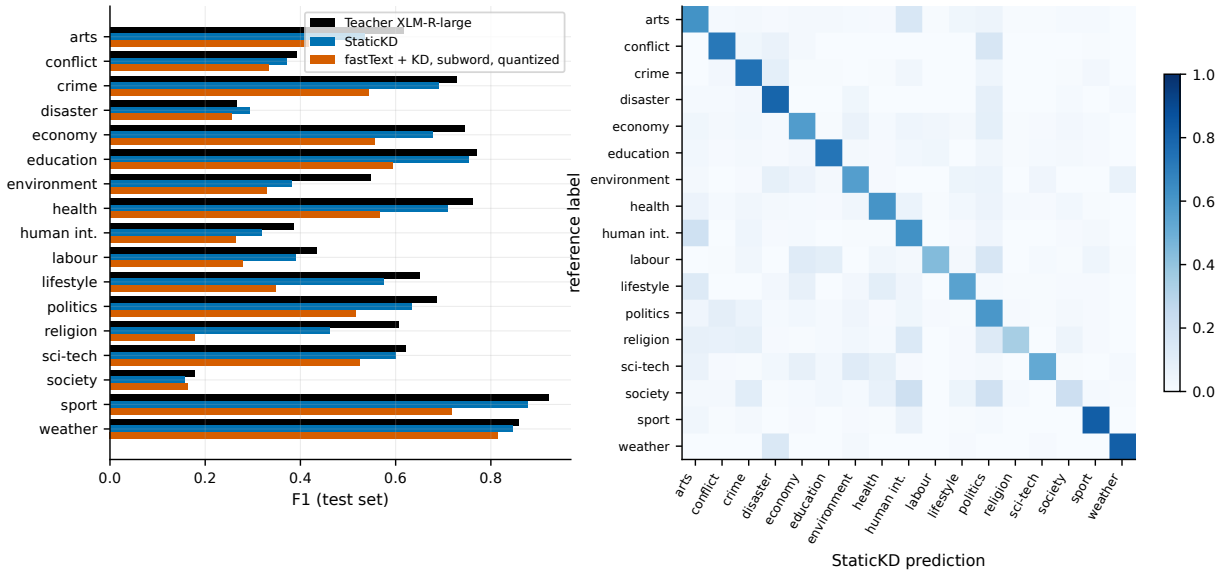


Fig. 9. F1 per label on the test set (left) and the StaticKD confusion matrix normalized per reference label (right).

to languages without data therefore relies on embeddings that training hardly changes. These embeddings lie in the same space as the trained ones, so the linear head learned on data-rich languages also applies to them.

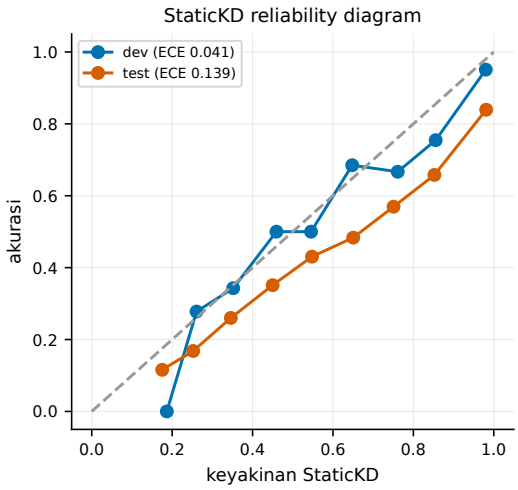


Fig. 10. Reliability diagram of StaticKD on the development and test sets.

The frozen-embedding ablation shows the other side of the same mechanism. Fine-tuning moves the embeddings of tokens from the training languages and improves the fit on these languages, which is why freezing lowers the development score. Tokens of other languages stay where potion put them, and the head, which has adapted to the moved embeddings, fits them less well. This is why frozen embeddings are better in languages with little or no distillation data. The final recipe keeps fine-tuning because the recipe was selected on a development set in the training languages, which cannot reveal this effect. For deployments that target languages without distillation data, frozen embeddings or a regularizer that keeps the embeddings close to potion are the safer choice, and both still fold into a table.

Soft labels, by contrast, add little or nothing. Hard teacher labels do not differ significantly from soft labels at $\tau = 1$. Raising the temperature to $\tau = 2$ lowers the held-out agreement without a significant change on the test set, and $\tau = 4$ lowers every metric (Section IV-C). This pattern fits the capacity-gap literature [25], [28]. A plausible explanation is that one linear layer over averaged embeddings cannot represent the inter-class structure in the tail of the teacher distribution (dark knowledge) [27]. A sharp distribution ($\tau = 1$) or hard labels give an attainable target, whereas a high τ asks the student to imitate a distribution shape beyond its capacity. For very-low-capacity students, effort is better spent on the coverage of the distillation data (languages, sites, topics) and on a multilingual initialization than on shaping the soft labels.

Per-language performance is driven mainly by how well a language is represented in the distillation and pre-training data, not by its script (Kruskal–Wallis $p = 0.065$). The weakest languages are African languages outside XLM-R pre-training that are also missing from the distillation data (ig, lg, ln, pcm, rn, sn, ti, yo). The teacher still reaches an accuracy of 0.643 on them, perhaps thanks to subwords shared with other languages and to sentence context, while StaticKD reaches only 0.407. The cheapest fix needs no change of architecture. News in these languages can simply be added to the distillation data and labeled by the teacher. The data ablation and the correlation between agreement and the amount of data ($\rho = 0.31$) support this step.

Agreement with the teacher does not always predict accuracy on editorial labels. Frozen embeddings lower the held-out agreement (-0.020) but raise the test macro-F1 (+0.014). Cutting the data to 130 thousand documents lowers the agreement (-0.009) without a significant change on the test set (-0.002). In line with [29], students should be selected with labels that do not come from the teacher and with a development set outside the training distribution, not with agreement with the teacher alone.

Test scores should be read according to the nature of the labels. An article about a criminal case involving an athlete can rightly sit in a sports section while a model labels it *crime*. Both answers are plausible under IPTC, which allows several topics per article. Comparisons between models remain fair because all models are scored against the same labels, and the model ranking does not change on the subset with capped site counts.

Compared with Sparse Distillation [23], StaticKD replaces an n-gram table trained from scratch with pre-trained multilingual embeddings folded into 17 logits per token. The model becomes far smaller and transfers to languages that are missing from the distillation data. Inference is the recurring cost in a news pipeline, since every incoming article is classified, and one million articles per day need 15.1 cores with the ONNX int8 teacher against 0.02 with StaticKD (Section IV-D).

For a real-time pipeline on servers without a GPU, StaticKD needs a 34 MB file and only `numpy` and `tokenizers` as dependencies. Its probabilities are better calibrated than those of the teacher (test ECE 0.139 against 0.245), which makes a cascade possible. With $\theta = 0.8$, the cascade accuracy (0.703) equals that of the teacher alone (0.699) at 1.92 articles per second per core, against 0.77 for the teacher alone. This mode suits batch jobs that need teacher-level accuracy. The compact variant lowers RAM but is only advisable when all target languages are covered by the distillation data.

VI. Limitations

Of the test articles, 47.5% carry JSON-LD section labels and 20.4% publisher categories. Section conventions differ between publishers, and one section reflects one editorial view of an article that may cover several topics. The mapping table is conservative, validated against the IPTC tree, and reviewed by the first author, but the articles were not re-annotated by humans. Absolute scores on these labels therefore measure agreement with the categorization of publishers, especially for labels with unclear boundaries (*society*, *human interest*, *lifestyle*).

MasakhaNEWS labels were assigned by human annotators under their own scheme and then mapped to IPTC. Only MN-DS is annotated directly with IPTC topics, and it is English only. The multilingual IPTC test set of Kuzman and Ljubešić [2] is not public, so it was not available for this study.

Label coverage per language is uneven, with a median of 7 labels per language, and the African languages cover only four

or five broad topics. Macro-F1 is computed over the labels present, so per-language scores are not directly comparable between languages with different label coverage. Publisher diversity is also limited, as 27 languages come from practically one publisher. These are BBC/VOA for the African languages, L3Cube for the Indian languages, and a single site for several CC-NEWS languages. Claims for these languages are limited to those publishers, and the site-cluster bootstrap widens their confidence intervals accordingly.

The teacher bounds the student from above, and its errors and biases are copied. The held-out set measures agreement with the teacher, not correctness.

The hyperparameters of fastText and TF-IDF + SVM were set once without a search, whereas the StaticKD recipe was selected from 72 runs on the development set. This unequal tuning budget may widen the accuracy gap to these two baselines. Two related methods were not run as baselines. Sparse Distillation [23] differs from fastText + KD and from the random-initialization ablation by the hidden layers of its deep averaging network, its soft-logit targets, and its billion-parameter n-gram table designed for English. The classifier trainer of Model2Vec [11] trains on hard labels, adds a 512-unit hidden layer, and learns a weight per token. The ablations test hard labels and a 512-unit hidden layer one at a time, and neither differs significantly from the final recipe on the development or test set (Table IX), but the two together and the learned token weights were not tested. A hidden layer would also prevent folding the model into one table. MiniLM-L6 and DistilmBERT were trained and evaluated with at most 256 tokens and therefore see less than half of most test articles, so their accuracy may be underestimated.

The benchmark ran on one machine in one session, with 30 latency documents for the teacher and the transformers. Absolute numbers on a server will differ and the latency sample of the heavy models is small, but ratios between models measured under the same protocol remain informative.

StaticKD is much less accurate on African languages that are absent from both the distillation data and pre-training (Section V). Used as a topic filter, it could misclassify or drop news in these languages more often. During deployment, accuracy should be monitored per language, and low-confidence articles should be routed to the teacher through the cascade mode.

The study covers only top-level IPTC labels and single-label classification. Deeper levels and multi-topic articles remain untested.

VII. Conclusion

The open multilingual IPTC classifier needs 1.2 to 3 seconds per article on one CPU core. StaticKD distills it into a table of 17 logits per token, initialized from multilingual static embeddings and trained on 285,999 teacher-labeled articles in 69 languages. On 12,744 test articles in 47 languages, with labels that do not come from the teacher, StaticKD reaches a macro-F1 of 0.545, or 91.1% of the teacher. It processes 473 articles per second per CPU core, about 616 times the speed of the ONNX int8 teacher, with a 34 MB model that needs no deep learning library.

TABLE XIV
Licenses of the Assets Used and Released

Asset	License
Teacher `classla/multilingual-IPTC-news-topic-classifier`	CC BY-SA 4.0
Embeddings `minishlab/potion-multilingual-128M`	MIT
MasakhaNEWS	CC BY-NC 4.0
MN-DS, L3Cube-IndicNews, IndoNews	CC BY 4.0
Bangla news [20]	not stated on the dataset card
EMMediaTopic 1.0	CC BY-SA 4.0
CC-News 2019–2021, CC-NEWS 2024–2026	Common Crawl terms of use
StaticKD code (release)	MIT
StaticKD models (release)	CC BY-SA 4.0 (inherited from the teacher)
Test-set manifest (release)	CC BY-SA 4.0

The accuracy of this very small student comes mainly from multilingual distillation data and from pre-trained multilingual embeddings, while soft labels are no better than hard labels. Fine-tuning the embeddings raises accuracy in the training languages but lowers it in languages without distillation data, whose embeddings training rarely or never updates. The test set can be reused to evaluate other IPTC classifiers.

Several directions follow from these results. The distillation data should be extended to the weak languages, especially the African languages outside XLM-R pre-training. A fine-tuning regularizer that keeps embeddings close to the pre-trained space could give in-distribution accuracy and cross-lingual transfer at the same time. Tokenization takes 98% of the inference time, so a faster tokenizer, for instance a direct word-to-table lookup, is the next source of speed. Any pruning of the vocabulary has to keep cross-lingual tokens. Cheap non-linear heads with quantized embeddings, hierarchical classification into IPTC level 2, multi-label classification, and human IPTC annotation of a multilingual sample of the test set remain open.

Code, Model, and Data Availability

The code for data collection, test-set construction, teacher labeling, training, evaluation, and benchmarking, together with a notebook that reproduces every number, table, and figure of this paper, is available at [TODO: GitHub URL and Zenodo DOI]. The StaticKD models are available at [TODO: Hugging Face URL]. The test set is released as a manifest of article URLs or IDs, labels, label origins, and text hashes, with a reconstruction script. Article texts are not redistributed because of publisher copyright and the licenses of the source datasets. The section-to-IPTC mapping table is released with the code. EMMediaTopic is available from the CLARIN.SI repository [9]. Table XIV lists the licenses of the assets used and released in this study.

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